Sound Immersion

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1. Introduction

Sound is vibrations emitted by a physical source, reaching the human ear, transformed into neural signals and interpreted by the brain within a given auditory context, often called an acoustic configuration. This process allows us reconstructing a detailed "**sound image**" that conveys not only speech and emotions but also spatial cues such as distance, direction, and environment.

1.1. The acoustic configuration

It determines the final perception of sound including several key components:

The geometry of the scene –

This refers to the positions, distances, and dimensions of the sound sources in the real or the virtual environment. Their spatial arrangement directly shapes how sound waves propagate to the human ear.

Acoustic and processing functions in the virtual world –

These functions simulate how sound is transformed when interacting with materials—such as absorption, reflection, diffraction, and occlusion—replicating the physical behavior of sound in real life.

Capture and playback devices –

In the physical world, sound is captured by the human ear; in digital environments, playback is ensured through headphones or speakers, which influence the quality, spatial cues, and the immersion level.

1.2. The interpretation

On the cognitive side, the interpretation refers to the mental reconstruction of the original scene based solely on the auditory perception—even when some elements or objects are hidden. The human brain infers distance, direction, movement, and even the nature of unseen objects through sound cues alone.

> 👉 When auditory perception is combined with vision, the sense of presence is greatly enhanced, allowing the body to feel fully immersed in the virtual world.

1.3. HMI functions

Another important function of sound is its role in Human–Machine Interaction (HMI). System feedback sounds—such as clicks, alerts, confirmations, or ambient cues—accompany the user's actions and increase usability, intuitiveness, and overall comfort.

2. 3D Sound

A 3D sound is a digitally processed audio signal that is designed to reproduce spatial cues allowing the listener to reconstruct a full three-dimensional scene from what he hears. When properly played back—typically through headphones or spatial audio speakers—This sound enables the user to perceive the position, distance, and movement of each sound source within the environment. In other words, the listener can localize where each sound is coming from as if it were occurring in the original real space.

3. Surround Sound (Multichannel Audio)

The surround sound—also known as **multichannel audio**—is one of the earliest and most widely adopted techniques for creating a 3D auditory experience. The principle is to distribute sound across multiple loudspeakers placed around the listener, thereby expanding the auditory field and enhancing immersion.

3.1. Principle and Objective

Surround sound relies on the idea that the sound can be spatialized by **physically placing multiple loudspeakers at different angles** and positions around the listener. Each speaker reproduces a distinct audio channel, allowing the auditory system to perceive directionality and movement through natural differences in –

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➔ **Time of arrival (interaural time differences)**
➔ **Intensity (interaural level differences)**
➔ **Spectral filtering (head and torso effects)**

3.2. Typical Configurations

Surround sound systems are commonly classified according to the number of channels and their arrangement. Here are some examples –

➔ **5.1** (Front Left, Front Right, Center, Surround Left, Surround Right, + subwoofer) – the most widely used in cinema and home theaters.
➔ **7.1** (adds Rear Left and Rear Right for increased back coverage).
➔ **9.1 or 11.1** (adds height channels for vertical dimension).

☞ The first number indicates the number of full-range channels, while the ".1" refers to a Low-Frequency Effects (LFE) channel dedicated to bass.

3.3. Rendering Process

In multichannel systems, spatialization occurs in two main stages:

a. Audio Encoding

The audio mix is created so that specific sounds are assigned to specific channels or are dynamically panned between the channels (e.g., moving car left to right is gradually shifted across multiple speakers).

b. Playback / Rendering

During playback, each audio channel is routed to its corresponding loudspeaker in the environment. The listener's perception of space emerges because the sound field surrounds them physically.

3.4. Strengths

➔ **Natural and Robust Spatialization**

Spatial cues are produced physically by real speakers → Perception of directionality is strong and stable.

➔ **High Quality for Cinematic Experiences**

It reproduces wide, dynamic scenes and convey movement, ambience and envelopment effectively.

➔ **Industry Standard**

Formats (**Dolby Digital**, **DTS**, **THX)** are adopted in film, broadcasting, home cinema, and gaming.

3.5. Limitations

➔ **Requires Physical Speaker Placement**

Accurate spatialization depends on correct speaker positioning → Less flexible than other methods.

➔ **Limited Personalization**

Does not adapt to head movements or individual ear shapes → Insufficient spatial accuracy for VR.

➔ **Not Suitable for Mobile or Head-Mounted Displays**

Surround sound requires a fixed listening position ("**sweet spot**"), however, in VR, users turn freely.

3.6. Use Cases

➔ Cinema and home theater systems;
➔ Video games on consoles or PCs;
➔ High-end presentation rooms and simulators;
➔ Theme parks and multi-projector spaces.

4. Principle of Loudspeaker Operation and Types of Speakers Drivers

4.1. 1. Basic Principle of Loudspeaker Functioning

A loudspeaker is a device that converts **electrical audio signals** into **mechanical vibrations** that generate **sound waves** in the air. This process relies on the interaction of three main components:

1. **Voice coil:** A coil of wire that receives the electrical signal.
2. **Magnet:** Creates a static magnetic field.
3. **Diaphragm (cone or dome):** Moves back and forth to push air and produce sound.

When the audio signal passes through the voice coil, it produces a varying magnetic field. This magnetic field interacts with the permanent magnet, causing the coil—and therefore diaphragm—to move. This movement displaces air at audible frequencies, creating the sound the listener perceives.

Because different frequencies require different types of movements and diaphragm sizes, no single speaker driver is capable to reproduce the entire audible spectrum (20 Hz – 20 kHz) with high fidelity. This leads to the use of different specialized drivers and multi-way speaker systems.

4.2. 2. Types of Speakers Drivers (According to Frequency Range)

Subwoofers – Lowest Frequencies (~20–120 Hz)

- ➔ **Size –** Very large drivers
- ➔ **Function –** Reproduce deep bass and sub-bass
- ➔ **Role –** Often used to cover the lowest frequencies for the entire system
- ➔ **Why separate?** Low frequencies require **large diaphragm excursions** and high power, which smaller drivers cannot handle.

Woofers – Low Frequencies (~40–1000 Hz)

- ➔ **Size –** Typically **10–38 cm** in diameter
- ➔ **Function –** Reproduce low to lower-mid frequencies
- ➔ **Limitations:**
  - ✓ Perform **poorly at high frequencies**
  - ✓ Produce **distortion** when pushed into high-frequency ranges
- ➔ **Use –** Essential in 2-way, 3-way, and 4-way systems
- ➔ Woofers cover part of the midrange in simpler designs.

Midrange Drivers – Middle Frequencies (~250–5000 Hz)

- ➔ **Size – 8–17 cm**
- ➔ **Function –** Reproduce the core of human hearing: speech, instruments, and most detailed sounds
- ➔ **Advantage –** Reduces strain on woofers and tweeters, improving clarity
- ➔ **Presence –** Found mainly in **3-way** and **4-way** systems

Tweeters – High Frequencies (~2000–20,000 Hz)

- ➔ **Size –** Small dome or cone drivers, **1–5 cm** in diameter
- ➔ **Function –** Reproduce high frequencies (treble) such as articulation, sibilance, and brightness
- ➔ **Important constraint:**
  - ✓ **Cannot handle low frequencies**
  - ✓ Sending bass or low midrange to a tweeter can **cause mechanical damage**

4.3. Multi-Way Speaker Systems

The used speaker systems combine different drivers to cover the full frequency spectrum with accuracy. These systems are classified according to the **number of driver types** they integrate:

2-Way Systems

- ➔ **Drivers**
  - ✓ Woofer (covers low + some midrange), Tweeter (high frequencies)
- ➔ **Advantages:** Simplicity, low cost
- ➔ **Limitation:** Less precise midrange reproduction

3-Way Systems

- ➔ **Drivers –** Woofer, Midrange, Tweeter
- ➔ **Advantages:**
  - ✓ Better fidelity
  - ✓ Precise distribution of roles
  - ✓ Reduced distortion
- ➔ **Ideal for:** Home audio, monitoring, cinema

4-Way Systems

- ➔ **Drivers –** Subwoofer, Woofer, Midrange, Tweeter
- ➔ **Advantages**
  - ✓ Optimal coverage of all frequency bands
  - ✓ High clarity and dynamic range
- ➔ **Use:** Professional audio, high-end installations

4.4. 4. The Crossover Principle

A **crossover** is an electronic filter that divides the audio signal into separate frequency bands and directs each band to the appropriate desired driver:

- ➔ **Low frequencies** → Subwoofer / Woofer
- ➔ **Mid frequencies** → Midrange driver
- ➔ **High frequencies** → Tweeter

Crossovers prevent:

➔ **Distortion** (e.g., a woofer trying to reproduce high frequencies)

➔ **Damage** (e.g., low frequencies reaching a tweeter)

Crossovers may be:

➔ **Passive** (integrated in the speaker)

➔ **Active** (external electronic filtering before amplification)

They ensure that each driver operates in its optimal range, maximizing audio quality.

Driver Type	Frequency Range	Size	Role	Notes
Subwoofer	20–120 Hz	Very large	Deep bass	System-wide low-frequency support
Woofer	40–1000 Hz	10–38 cm	Low/mid	Distortion at high frequencies
Midrange	250–5000 Hz	8–17 cm	Speech + core details	Improves clarity
Tweeter	2000–20,000 Hz	1–5 cm	High frequencies	Cannot handle bass

5. Techniques for Reproducing Real 3D Sound

Reproducing a realistic three-dimensional auditory scene requires audio technologies designed to create spatial cues in all the perceived directions—horizontally, vertically, and with depth. Modern 3D audio techniques go far beyond traditional stereo or surround sound by incorporating **height channels**, **full-sphere spatialization**, and **object-based rendering** that adapts to listener movement.

5.1. Reproducing Real 3D Sources with Loudspeakers (Hi-Fi Multidirectional Systems)

Traditional high-fidelity (Hi-Fi) loudspeaker systems reproduce sound sources as accurately as possible within a room creating a wide stereo image or a horizontal surround field. However, to achieve a more immersive 3D auditory experience, the system can be extended in two ways:

Adding Height Information

Height cues improve vertical localization, enabling listeners to perceive sounds above and below them. Two architectural strategies exist:

The first one – Ceiling Speakers

➔ Physically installed above the listener.

➔ Deliver direct height information.

➔ Provide reliable vertical cues / Require complex room infrastructure.

➔ Common in **professional theaters**, **immersive rooms**, and **advanced home theaters**.

The second one – Upward-Firing (Dolby Atmos) Modules

➔ Loudspeakers angled upward to **reflect** sound off the ceiling.

➔ The listener perceives the reflected wave as coming from above.

➔ Easier to install / Less accurate than dedicated ceiling speakers.

➔ Widely adopted in home Atmos systems.

This extends auditory field from **2D surround plane** to **3D hemispherical or full-sphere configuration**.

5.2. Channel-Based 3D Audio

Channel-based audio defines sound spatialization by assigning audio signals to specific loudspeaker channels. The motion or direction perception are created by **panning** sounds between these channels.

➔ **Principle**

Each loudspeaker corresponds to a fixed direction in space. The listener perceives a sound's position depending on how the signal is distributed across these channels.

➔ **Advantages**

- ✓ Simple, stable, and compatible with existing speaker layouts.
- ✓ Effective for cinema, home theater, and traditional VR caves (CAVEs).

➔ **Limitations**

- Not adaptive: the experience depends on the listener being in the "**sweet spot**".
- Does not natively encode vertical or continuous spatial information.
- Lacks accuracy when the listener moves or rotates in VR.

This format is used in systems such as **5.1, 7.1, 9.1, and Atmos channel-based configurations**.

5.3. Ambisonics – Full-Sphere Spatial Audio

Ambisonicsis a **full-sphere audio format** capable of reproducing sound from **all the directions**, including height and depth components too. Ambisonics encodes the **sound field itself**, not the loudspeaker layout.

➔ **Principles**

Ambisonics represents the sound field using mathematical components:

- ✓ Omnidirectional component (W)
- ✓ Directional components (X, Y, Z)

Higher-order ambisonics adds more components for increased spatial resolution.

➔ **Orders**

First-Order Ambisonics (FOA) – Lower precision; commonly used in consumer VR. (YouTube VR)
Higher-Order Ambisonics (HOA) – Increases accuracy, spatial resolution, and realism (Facebook 360).

➔ **Advantages**

- ✓ Full-sphere representation.
- ✓ Adaptable to any configuration of loudspeakers or headphones.
- ✓ Highly suitable for **3D video**, **360° VR environments**, and **immersive rooms (CAVEs)**.

➔ **Limitations**

- ✓ Requires decoding and potentially large numbers of speakers to get optimal results.
- ✓ Capturing high-order ambisonics requires specialized non-standard microphones.

Ambisonics represents one of the most flexible and powerful 3D audio formats in VR.

5.4. Object-Based Audio

Object-based audio represents a major evolution from channel-based systems in which, instead of assigning sound to channels, each sound is treated as an **audio object** with metadata that describes: its position in 3D space, trajectory and movement, and its acoustic characteristics.

During playback, the rendering engine calculates how each object should be reproduced depending on:

- ✓ Listener orientation
- ✓ Speaker configuration
- ✓ Acoustic simulation (room, materials)
- ✓ Device type (headphones, multichannel speakers, VR headset)

➔ **Advantages**

- ✓ Extremely flexible – adapts to any playback system.
- ✓ High precision, ideal for VR and interactive environments.
- ✓ Supports dynamic scenes where sound sources move or the listener rotates.

➔ **Examples**

- ✓ Dolby Atmos (object + channel hybrid)
- ✓ DTS:X
- ✓ Game audio engines (Unity Spatializer, Oculus Audio, Steam Audio)

6. Methods of Sound Recording: Mono, Stereo, and Binaural

Sound recording techniques determine the amount of spatial information preserved when capturing an acoustic scene. While all methods store the main part of the audio information, they differ significantly in their ability to reproduce the corresponding realism, depth, and spatial localization. The three fundamental approaches are **Mono**, **Stereo**, and **Binaural recording**, each serving specific goals and constraints:

6.1. Mono Recording

Mono (monophonic) recording uses **single microphone** to capture sound. Regardless of the number of output channels (even if exported as "stereo"), the signal originates from **one single point**. If a mono signal is played through two output channels (left and right):

- ➔ The **same waveform** is sent to both the speakers.
- ➔ **No time delay** between channels.
- ➔ **No amplitude difference** between channels.

Mono lacks spatial cues needed for realistic auditory localization because:

- ➔ Only **one channel** of information exists.
- ➔ There is **no interaural time difference (ITD)**: sound reaches both ears at exactly the same time.
- ➔ There is **no interaural level difference (ILD)**: both listener ears receive the same intensity.
- ➔ There is no directional filtering (HRTF) encoded.

As a result, the listener perceives the sound as coming from a **centered, undefined** position. Mono recordings are therefore **not spatial** at all but are useful for –

- ➔ Voice recording (podcasts, telephony)

- ➔ Archival purposes
- ➔ Environments where bandwidth is limited
- ➔ Applications where spatial realism is unnecessary

6.2. Stereo Recording

Stereo (two-channel) recording recreate a **horizontal sound image** by using two microphones placed in a specific configuration allowing the listener to perceive width, direction, and depth on a horizontal plane. Stereo technique uses **two channels** (Left and Right) carrying **similar but not identical** signals. Localization relies on differences between the channels:

- ➔ **Interaural Time Differences (ITD)** – Slight time offset between the two microphones
- ➔ **Interaural Level Differences (ILD)** - One microphone receives louder signal depending on direction

Two major techniques are commonly used to record stereo sounds:

XY Technique (Coincident Pair)

- ➔ Two directional microphones placed at the **same point** with an angle between them.
- ➔ Ideal for **indoor** and **static** sources.
- ➔ Minimal phase problems because capsules are nearly coincident.
- ➔ Creates a clean, stable stereo image.

ORTF Technique (Near-Coincident Pair)

- ➔ Microphones separated by about 17 cm, angled at 110°.
- ➔ Mimics human ear spacing.
- ➔ Better for **outdoor**, **dynamic**, or **naturalistic** recordings.
- ➔ Produces wider and more realistic stereo images.

What Is a “Stereo Sound Image”?

It is the virtual horizontal space perceived between two speakers or headphones including:

- ➔ **Left–right positioning** of sources
- ➔ **Apparent width** of instruments or environment
- ➔ **Depth** resulting from relative loudness and reverberation

Stereo does not reproduce height/full 3D space but provides convincing two-dimensional auditory scene.

6.3. Binaural Recording

This recording aims to reproduce sound **exactly as a human hears it**, using **dummy head** or miniature microphones inserted into a listener’s ears capturing all the acoustic cues that allow spatial localization. The core concept is **Head-Related Transfer Function – HRTF** (**Anatomical Transfer Function – ATF)**.

Head-Related Transfer Function (HRTF)

HRTF describes how sound is filtered by the **anatomy of the listener** before reaching eardrum including:

- ➔ Shape and size of the **head**
- ➔ Structure of the **outer ear (pinna)**
- ➔ Geometry of the **ear canal**
- ➔ **Nasal and oral cavities**
- ➔ Tissue density and skull reflections

Each of above elements **boosts some frequencies** and **attenuates others**, depending on:

- ➔ The **direction** of the sound source
- ➔ The **distance** of the source
- ➔ Whether the source is above, behind, or below the listener

This filtering provides critical cues for **3D localization**.

What Binaural Recording Enables

When replayed through headphones, binaural recordings can reproduce:

- ➔ Full 3D space (front, back, above, below)
- ➔ Distance perception
- ➔ Elevation cues
- ➔ Precise horizontal and vertical localization
- ➔ Occlusion and filtering by head and torso
- ➔ Immersive realism similar to natural listening

Because it encodes the ear’s natural acoustic modifications, binaural audio is considered ideal for:

- ➔ Virtual Reality (VR) and Augmented Reality (AR)
- ➔ 3D audio simulations

- ➔ Ambisonics decoding
- ➔ Spatialized game audio
- ➔ Research on perception and auditory cognition

Limitations

- ➔ Best experienced with **headphones** (loudspeakers reduce accuracy)
- ➔ HRTF varies across individuals—generic HRTFs may not perfectly match every listener.

7. Sound Rendering Pipeline in VR

In VR environments, sound rendering follows a structured pipeline that transforms raw the audio signals into a final spatially coherent and immersive auditory experiences. This pipeline ensures that each sound source is correctly generated, positioned, processed, and rendered according to the listener's position and the acoustic properties of the virtual scene content:

7.1. Generation

Each sound is generated / associated with a **specific object**, treated as **isolated from the environment**. This step defines the source audio signal before spatial or environmental effects are applied.

7.2. Association

The generated sound is then linked to the **object's movement and position** within the virtual scene. At this stage, **distance, relative orientation, and environmental properties** (room acoustics, absorption, and reflections) are considered to model how the sound propagates over the scene.

7.3. Convolution / Spatialization

The system calculates the **acoustic transformations** required for each sound source to reflect the environment accurately including:

- ➔ **Convolutions** with impulse responses of virtual spaces
- ➔ **HRTF filtering** for binaural rendering
- ➔ **Occlusion and reflection effects**

7.4. Instantiation / Playback

Finally, the processed sound is **instantiated** in the scene, ready to be rendered through the playback system (the devices: headphones or speakers). This step ensures that the listener perceives the sound in the predefined correct **position, direction, and spatial context**.

8. 3D Sound Imaging

The 3D sound imaging process relies on a sequence of spatial analysis, signal processing, and rendering steps that collectively ensure accurate and immersive auditory perception in the virtual environment:

8.1. Spatial Analysis

The system computes **line-of-sight information** between the virtual listener and all sound sources in the scene to determine which sounds are directly audible, which are occluded, and the relative distances and angles for accurate the spatial rendering.

8.2. 2. Sound Processing

Each sound source is processed according to its **position and environment** using **digital signal processing (DSP) techniques**:

- ➔ **Delay –** Simulating distance and propagation time
- ➔ **Filtering –** Modeling the attenuation and the frequency-dependent absorption by the materials
- ➔ **Panning –** Positioning sounds in the stereo or multichannel field
- ➔ **Reverb and reflections –** Simulating environmental acoustics

These processes allow the listener to perceive both the **direction** and the **distance**, enhancing immersion.

8.3. 3. Output Rendering

The processed audio is rendered for **headphone playback**, generating a **stereo or binaural output** adapting dynamically to the user's position and orientation. In addition, using **head tracking**, the spatial cues are updated in real time, ensuring that sounds appear to originate from the correct locations.